Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title: Customer Analysis Dashboard using Power BI

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Customer Analysis Dashboard using Power BI

GitHub Repository Link: *https://github.com/nidhidev2122/Business_analytics*

Problem Statement

A company wants to understand its customers and sales performance. Using the order and customer data, build an interactive dashboard in Power BI that represents sales, revenue and profit with appropriate visuals, and answers the following business questions:

- Which product category is most popular?
- Which payment method is most frequently used?
- How does revenue change month by month?
- Which customers have made the most purchases?
- Are a small number of customers responsible for a large proportion of revenue?

Dataset Used

- **Orders table (5,000 rows):** OrderID, CustomerID, OrderDate, ProductCategory, Product, Quantity, UnitPrice, DiscountPercent, SalesAmount, PaymentMethod
- **Customers table (1,000 rows):** CustomerID, CustomerName, Gender, Age, City, State, IncomeBand, SignupDate, CustomerSegment
- **Files:** customer_analysis_orders_5000.csv and customer_analysis_customers_1000.csv
- **Tool used:** Microsoft Power BI Desktop (Power Query and DAX)

Steps Followed

Step 1: Data Collection

The two CSV files containing order transactions and customer details were used for this experiment. The Orders table holds one row per order, while the Customers table holds one row per customer. The two tables are linked by the CustomerID column.

Step 2: Data Import

Both CSV files were loaded into Power BI using Get Data → Text/CSV. The Power Query Editor was opened to inspect the columns and data types before building the model.

Step 3: Data Preparation

3.1 Checking Data Quality

Column Quality and Column Distribution were used to check both tables. No missing values were found in either table, and all 5,000 OrderIDs were unique, so no rows had to be removed. The orders cover 1 January 2025 to 30 December 2025 and belong to 916 of the

1,000 customers.

3.2 Correcting Data Types

- OrderDate and SignupDate: converted from Text to Date
- Quantity, UnitPrice and DiscountPercent: set as Whole Number
- SalesAmount: set as Fixed Decimal Number
- Text columns (CustomerID, ProductCategory, PaymentMethod, etc.): kept as Text

3.3 Creating Date Columns

Month, Month Number and Year columns were derived from OrderDate so that revenue could be analysed month by month and sorted in calendar order.

Step 4: Data Modelling

A one-to-many relationship was created in Model view from Customers[CustomerID] to Orders[CustomerID] (single direction). This allows customer attributes such as State, City and Customer Segment to filter the order data.

Step 5: Creating DAX Measures

The following measures were created for the KPI cards and charts:

```
Total Revenue = SUM(Orders[SalesAmount])
Total Orders = COUNTROWS(Orders)
Total Customers = DISTINCTCOUNT(Orders[CustomerID])
Total Quantity = SUM(Orders[Quantity])
Avg Order Value = DIVIDE([Total Revenue], [Total Orders])
```

Step 6: Building the Dashboard

KPI cards were placed at the top of the report page, followed by one visual for each business question. Slicers for Year, Product Category, State and Customer Segment were added so that the whole page can be filtered interactively.

No.	Business Question	Visual Used
1	Most popular product category	Bar chart – number of orders by ProductCategory
2	Most frequently used payment method	Donut chart – order count by PaymentMethod
3	Month-by-month revenue	Line chart – Total Revenue by Month
4	Customers with most purchases	Top 10 bar chart / table – orders per customer
5	Revenue concentration	Pareto chart – revenue per customer with cumulative %

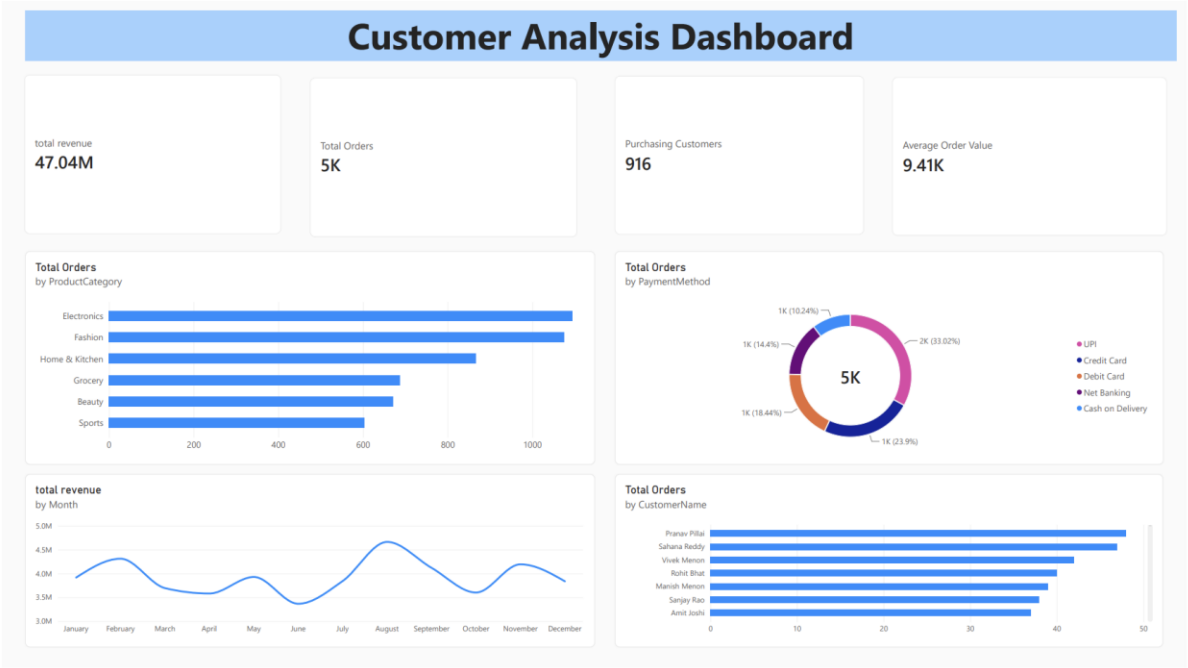


Figure 5: Customer Analysis Dashboard

Dashboard Analysis

Overall Performance (KPI Cards)

Metric	Value
Total Revenue	₹4.70 crore (₹47,039,602)
Total Orders	5,000
Total Quantity Sold	7,172 units
Active Customers	916 (of 1,000)
Average Order Value	₹9,408
Estimated Profit (20% assumed margin)	₹94.1 lakh (₹9,407,920)

Q1. Which product category is most popular?

Electronics is the most popular category with 1,095 orders, followed closely by Fashion with 1,076 orders. Home & Kitchen (867), Grocery (687), Beauty (672) and Sports (603) follow. Electronics also generates by far the highest revenue – about ₹3.75 crore, which is roughly 80% of the total – because of its high unit prices, whereas Grocery has the lowest revenue of about ₹5.4 lakh.

Q2. Which payment method is most frequently used?

UPI is the most frequently used payment method with 1,651 orders (33.0%), followed by Credit Card (1,195 orders, 23.9%), Debit Card (922, 18.4%), Net Banking (720, 14.4%) and Cash on Delivery (512, 10.2%). Digital payments therefore account for almost 90% of all orders.

Q3. How does revenue change month by month?

Monthly revenue stays within a fairly narrow band of about ₹33 lakh to ₹47 lakh through 2025, with no strong long-term upward or downward trend. The highest revenue was in August (₹46.6 lakh), followed by February (₹43.1 lakh) and November (₹41.9 lakh). The lowest was in June (₹33.7 lakh). Revenue dipped in March–April and again in June before recovering in July–August.

Q4. Which customers have made the most purchases?

The customers with the highest number of orders are listed below. C0553 placed the most orders (39), and C0553 and C0877 also generated the highest revenue among the frequent buyers (about ₹6.4 lakh each).

Customer ID	Number of Orders	Revenue (₹)
C0553	39	641,738
C0327	37	278,589
C0877	34	640,263
C0378	33	241,918
C0526	31	207,979

Q5. Are a small number of customers responsible for a large proportion of revenue?

Yes. Revenue is clearly concentrated among a minority of customers:

- The top 100 customers (about 11% of the 916 buyers) generate 44.1% of total revenue.
- The top 20% of customers (183) generate 62.4% of total revenue.
- About 311 customers (roughly 34% of buyers) account for 80% of revenue.

The concentration is not as extreme as the classic 80/20 rule, but a relatively small group of high-spending customers clearly drives a large share of revenue.

Summary

- **Orders table:** 5,000 unique orders with no missing values; dates, quantities and amounts converted to correct data types.
- **Customers table:** 1,000 customers; linked to Orders through CustomerID with a one-to-many relationship.

- **Measures:** Revenue, Orders, Customers, Quantity, Average Order Value and Estimated Profit created using DAX.
- **Visuals:** KPI cards, bar chart, donut chart, line chart, Top-10 chart and Pareto chart, with slicers for interactive filtering.

Conclusion

This experiment provided practical experience in building an interactive business dashboard in Power BI. The order and customer data were imported, checked, modelled with a relationship and analysed using DAX measures. The final dashboard answers all five business questions and shows that Electronics is the leading category, UPI is the preferred payment method, monthly revenue is relatively stable with a peak in August, and a small group of customers contributes a large share of total revenue.

Business Insights

- **Electronics drives revenue:** It makes up only about 22% of orders but around 80% of revenue, so stock availability and pricing in this category matter most.
- **Digital payments dominate:** UPI and cards cover most orders; promotions tied to UPI could be effective, and Cash on Delivery is only about 10% of orders.
- **Revenue is stable but seasonal dips exist:** March–April and June are weaker months and could be targeted with campaigns, while August performance can be studied and repeated.
- **Loyal customers matter:** The top 20% of customers generate over 60% of revenue, so a loyalty or rewards programme for high-value customers could protect a major revenue source.
- **Geography:** Karnataka (₹1.32 crore) and Maharashtra (₹95.7 lakh) are the top states by revenue.
- **Customer segments:** High Value customers contribute about ₹2.94 crore (63%) of total revenue.